Topic 12 – Magnetism and the motor effect

Students should:		Maths skills
12.1	Recall that unlike magnetic poles attract and like magnetic poles repel	
12.2	Describe the uses of permanent and temporary magnetic materials including cobalt, steel, iron and nickel	
12.3	Explain the difference between permanent and induced magnets	
12.4	Describe the shape and direction of the magnetic field around bar magnets and for a uniform field, and relate the strength of the field to the concentration of lines	5b
12.5	Describe the use of plotting compasses to show the shape and direction of the field of a magnet and the Earth's magnetic field	5b
12.6	Explain how the behaviour of a magnetic compass is related to evidence that the core of the Earth must be magnetic	5b
12.7	Describe how to show that a current can create a magnetic effect around a long straight conductor, describing the shape of the magnetic field produced and relating the direction of the magnetic field to the direction of the current	5b
12.8	Recall that the strength of the field depends on the size of the current and the distance from the long straight conductor	
12.9	Explain how inside a solenoid (an example of an electromagnet) the fields from individual coils a add together to form a very strong almost uniform field along the centre of the solenoid b cancel to give a weaker field outside the solenoid	5b
12.10	**Recall that a current carrying conductor placed near a magnet experiences a force and that an equal and opposite force acts on the magnet**	5b
12.11	**Explain that magnetic forces are due to interactions between magnetic fields**	
12.12	**Recall and use Fleming's left-hand rule to represent the relative directions of the force, the current and the magnetic field for cases where they are mutually perpendicular**	5b

Students should:		Maths skills
12.13	**Use the equation:** **force on a conductor at right angles to a magnetic field carrying a current (newton, N) = magnetic flux density (tesla, T or newton per ampere metre, N/A m) × current (ampere, A) × length (metre, m)** $\boldsymbol{F = B \times I \times l}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
12.14P	**Explain how the force on a conductor in a magnetic field is used to cause rotation in electric motors**	5b

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).

Suggested practicals

- Construct an electric motor.